Curium

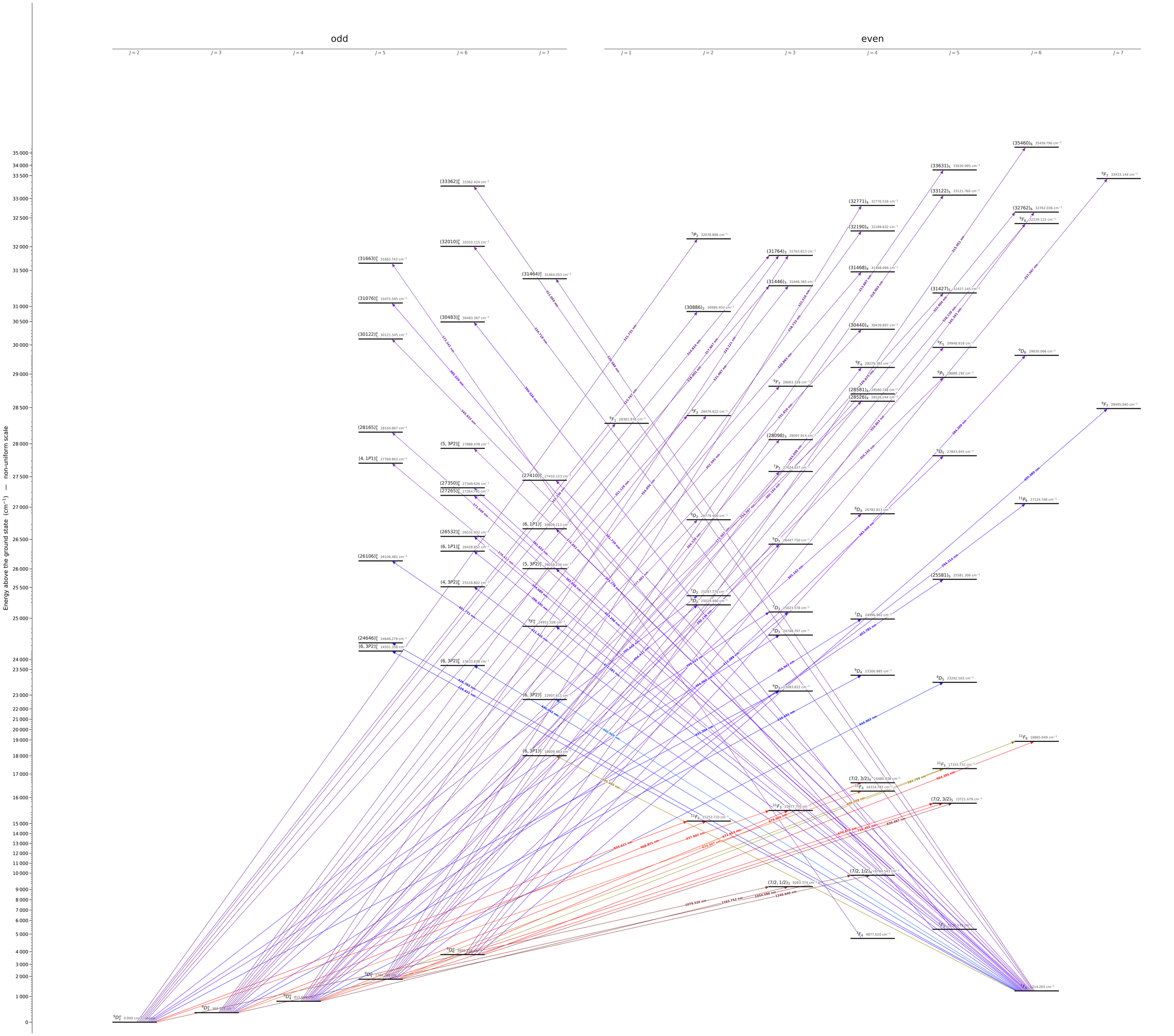
Cm (Cm I). Classified transitions below 2 µm (levels and lines from the literature): 83 levels, 88 lines.

Cm^+ ionisation limit 48330.68 cm^{-1} = 5.99224 eV (206.908 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| er \| J' \rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (from the cited literature; NIST lists no excited levels) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)

300 400 500 600 700 800 1000 1200 1500 2000



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	\|⟨J‖er‖J'⟩\| (ea₀)	A (s⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
$^9D^o_2$ – $^7D_2$	395.4481	395.3362	758.1083	–	–	–	–	first step of Mainz scheme C
$^9D^o_4$ – (25581)$_5$	403.7851	403.6710	742.4555	–	–	–	–	excitation from the thermally populated 9D4 level seen in…
$^9D^o_2$ – $^9D_3$	404.0600	403.9459	741.9503	–	–	–	–	first step of Mainz scheme B
$^9D^o_3$ – $^9D_2$	404.4112	404.2970	741.3060	–	–	–	–	excitation from the thermally populated 9D3 level seen in…
$^9D^o_3$ – $^7D_4$	404.9434	404.8290	740.3318	–	–	–	–	excitation from the thermally populated 9D3 level seen in…
$^9D^o_2$ – $^9D_3$	433.2038	433.0820	692.0356	–	–	–	–	first step of Mainz two-step resonance-ionization scheme…

λ, ν from the level energies of the cited literature (NIST lists no excited levels) unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$^7F_6$ – (33362)$^o_6$	311.059	246-244	6446.00
$^9D^o_2$ – $^5P_2$	311.731	246-244	-14210.00
$^9D^o_2$ – (33631)$_5$	313.807	246-244	-1199.00
$^9D^o_2$ – (31764)$_3$	314.824	246-244	-10073.00
$^9D^o_2$ – (35460)$_6$	315.951	246-244	-11812.00
$^9D^o_2$ – (31764)$_3$	317.847	246-244	-10103.00
$^9D^o_2$ – (31446)$_3$	318.002	246-244	-10403.00
$^9D^o_2$ – (32190)$_4$	318.733	246-244	-6595.00
$^9D^o_2$ – (33122)$_5$	318.903	246-244	-7675.00
$^9D^o_2$ – (31468)$_4$	320.863	246-244	-11392.00
$^9D^o_2$ – (31446)$_3$	321.087	246-244	-10433.00
$^9D^o_2$ – (32771)$_4$	322.516	246-244	-12052.00
$^9D^o_2$ – (32762)$_6$	322.604	246-244	-10613.00
$^9D^o_2$ – (31764)$_3$	323.121	246-244	-10133.00
$^9D^o_2$ – (30886)$_2$	323.767	246-244	-13431.00
$^7F_6$ – (32010)$^o_6$	324.718	246-244	4317.00
$^9D^o_2$ – (31427)$_5$	326.675	246-244	-10972.00
$^9D^o_2$ – $^9F_6$	328.139	246-244	-10193.00
$^7F_6$ – (31464)$^o_7$	330.580	246-244	4647.00
$^9D^o_2$ – (30440)$_4$	331.810	246-244	-12771.00
$^9D^o_2$ – $^9F_7$	337.567	246-244	-11272.00
$^9D^o_2$ – (32762)$_6$	345.391	246-244	-10852.00

Shift = ν(first isotope) − ν(second isotope).